

WEEK 4 – ASSIGNMENT

Objective:

Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory. Steps: Explore Azure Portal and create a Resource Group. Create a Storage Account, Blob Container, and upload a CSV file. Create Azure Data Factory and explore ADF UI (Author, Monitor, Manage). Create Linked Service (Blob Storage) and datasets (source and destination). Use Get Metadata activity to retrieve file information. Build a pipeline using Copy Data activity and configure source and destination. Execute the pipeline using Debug/Trigger and monitor execution. Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage. Implement an end-to-end pipeline (Blob → ADF → Destination) with metadata validation. Output: Screenshots (Resource Group, Storage, ADF, Pipeline) + pipeline execution results + brief summary.

TASK 1: Create a Resource Group

Screenshot : Creating the Resource Group " RG-Celebal-Week4"

Microsoft Azure

Search resources, services, and docs (G + I)

Copilot

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CVR COLLEGE OF ENGINEERING L

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

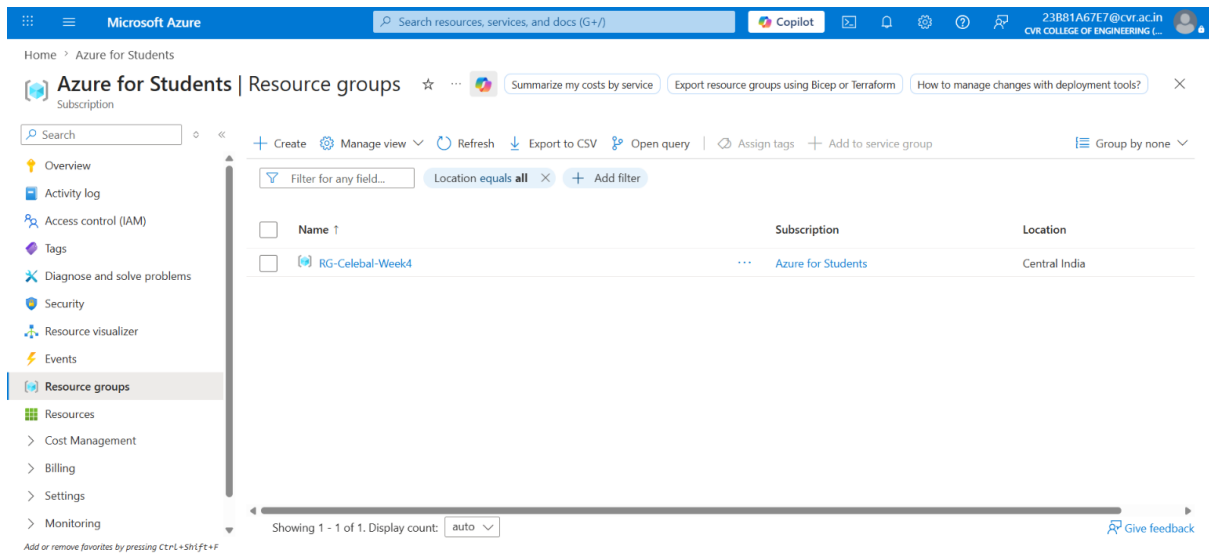
Subscription *

Resource group name *

Region *

Previous Next Review + create

Screenshot : Successfully Created Resource Group (RG-Celebal-Week4)



TASK 2 :

- Create a Storage Account

Screenshot : Creating the Storage Account " adfcelebalstorage."

Home > Storage center | Blob Storage

Create a storage account

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure for Students

Resource group * RG-Celebal-Week4

Create new

Instance details

Storage account name * adfcelebalstorage

Region * (Asia Pacific) Central India

Deploy to an Azure Extended Zone

Preferred storage type Choose preferred storage type

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more

Performance * Standard: Recommended for most scenarios (general-purpose v2 account)

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Give feedback

Screenshot : Successfully Created Storage Account " adfcelebaladfstorage."

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > adfcelebalstorage_1782043505175 | Overview

adfcelebalstorage

Storage account

Create an LCM rule for my storage account

Help me connect my application to this storage account.

Help me save costs on my storage

Search

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CLI / PS

Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Essentials

Resource group (move)

Location

Primary/Secondary L...

Subscription (move)

Subscription ID

Disk state

Tags (edit)

Performance

Replication

Account kind

Provisioning state

Created

Properties

Monitoring

Capabilities (7)

Recommendations (0)

Tutorials

Tools + SDKs

Blob service

Hierarchical namespace

Default access tier

Blob anonymous access

Blob soft delete

Continuous availability

Disabled

Hot

Disabled

Enabled (7 days)

Enabled (7 days)

Security

Require secure transfer for REST API operations

Storage account key access

Minimum TLS version

Infrastructure encryption

Enabled

Enabled

Version 1.2

Disabled

- Create a Blob Container

Screenshot : Creating the Blob Container " source-data."

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > adfcelebalstorage_1782043505175 | Overview > adfcelebalstorage

adfcelebalstorage

Storage account

Containers

Search

Add container

Upload

Refresh

Delete

Change access level

Restore containers

Search containers by prefix

Showing all 1 items

	Name	Last modified	Anonymous access level
☐	source-data	6/21/2026, 5:35:49 PM	Private

New container

Name *

source-data

Anonymous access level

Private (no anonymous access)

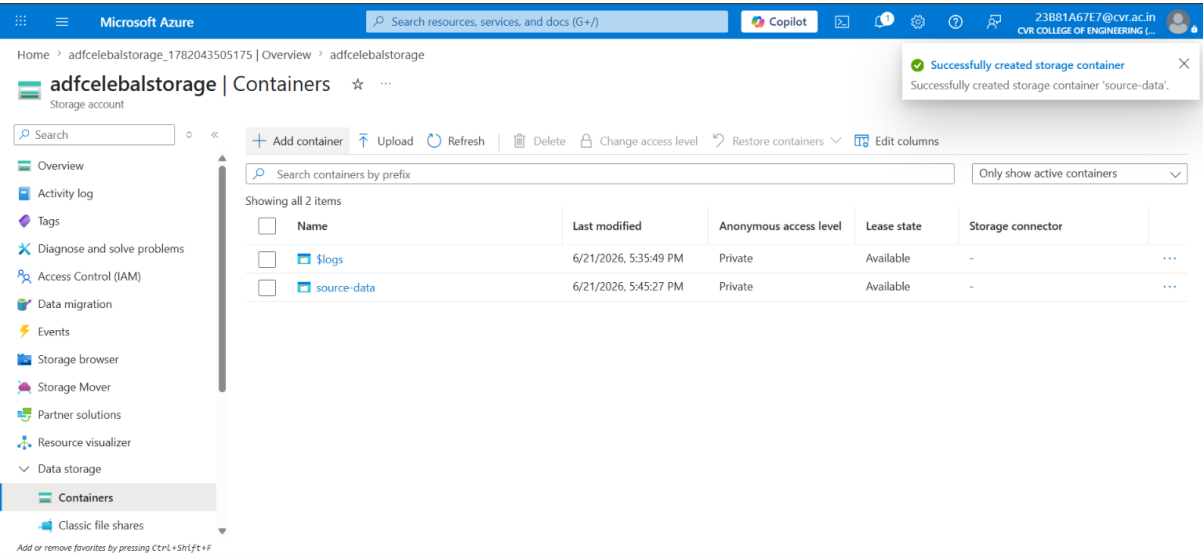
The access level is set to private because anonymous access is disabled on this storage account.

Advanced

Create

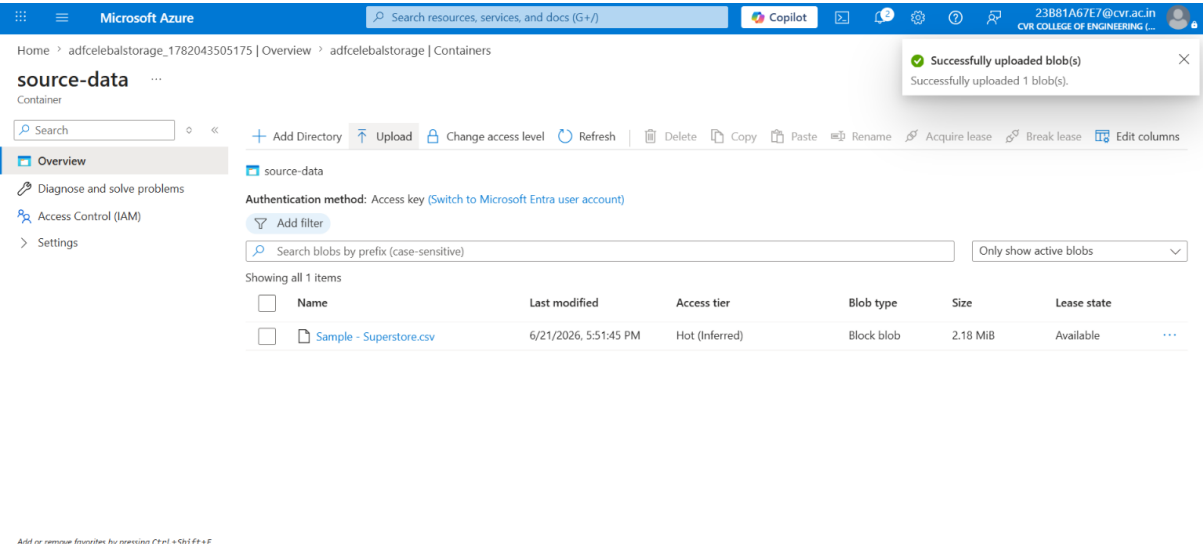
Give feedback

Screenshot : Successfully Created Blob Container " source-data."



- Upload a CSV file

Screenshot : Successfully Uploaded Sample-Superstore CSV File



TASK 3 :

- **Create Azure Data Factory**

Screenshot : Creating the Azure Data Factory “ADFCelebalWeek4”

Microsoft Azure

Home > Create a resource > Marketplace

Create Data Factory

Basics | Git configuration | Networking | Advanced | Tags | Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *
[Create new](#)

Instance details

Name *

Region *

Version *

Upgrade message: We recommend you **upgrade to Fabric Data Factory** for a modern, unified data integration experience. [Try Fabric](#) instead.

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

Screenshot : Successfully Created Azure Data Factory “ADFCelebalWeek4”

Home > Microsoft.DataFactory-20260621180447 | Overview > RG-Celebal-Week4

ADFCelebalWeek4

Data factory (V2)

Search Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Getting started
- Monitoring
- Automation
- Help

Essentials

Resource group (move) : [RG-Celebal-Week4](#)

Status : Succeeded

Location : Central India

Subscription (move) : [Azure for Students](#)

Subscription ID : 96058607-2e8e-4ee3-9a41-1a34012b5c49

Type : Data factory (V2)

Getting started : [Quick start](#)

Azure Data Factory Studio

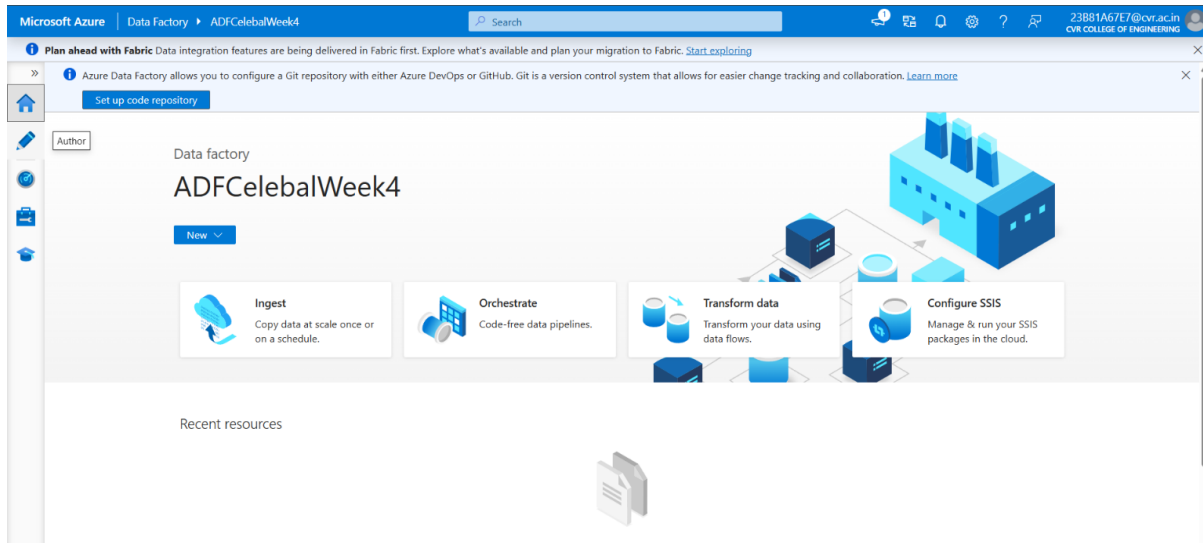
[Launch studio](#)

[Quick Starts](#) [Tutorials](#) [Template Gallery](#) [Training Modules](#)

Add or remove favorites by pressing Ctrl+Shift+F

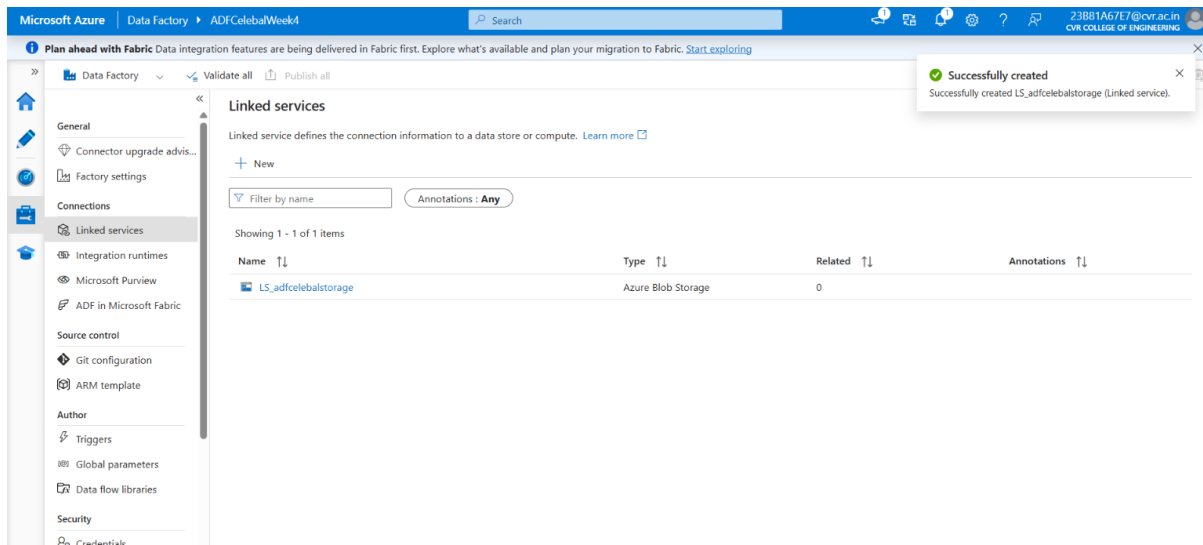
- Explore ADF UI (Author, Monitor, Manage).

Screenshot : ADF Studio Home Page.



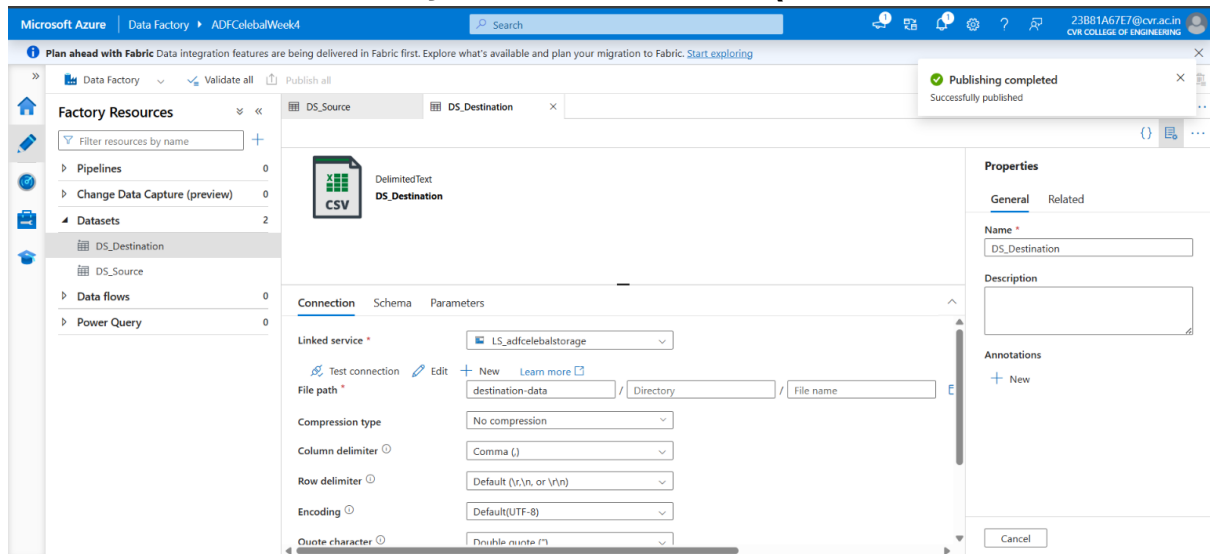
TASK 4 :

- Create Linked Service (Blob Storage)



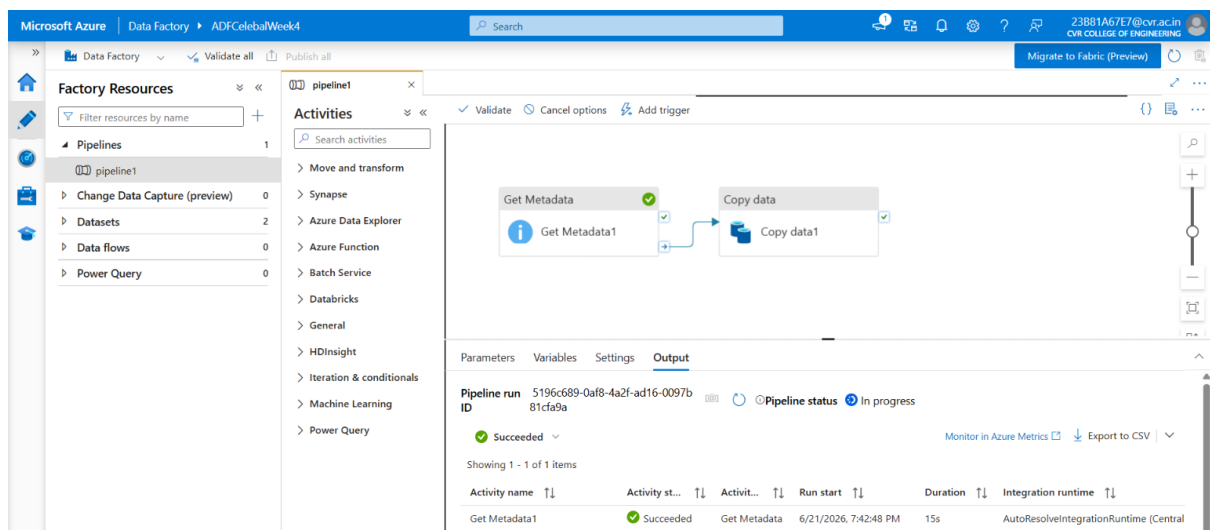
- Create datasets (source + destination)

Screenshot : Successfully Created datasets (source + destination)

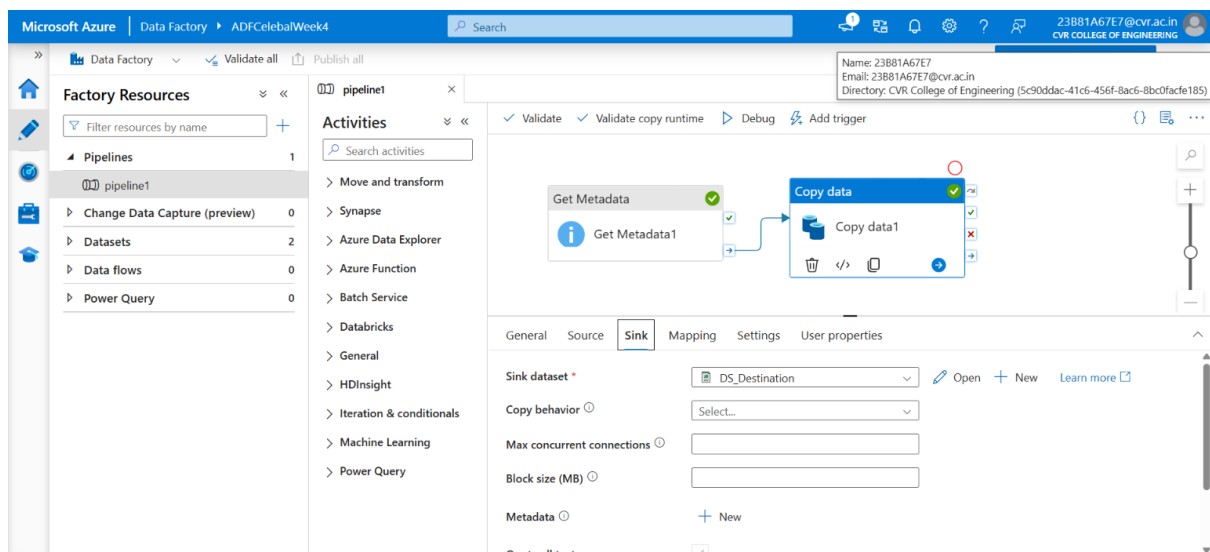


TASK 5 : Use Get Metadata activity to retrieve file information.

Screenshot : Successful Execution of Get Metadata Activity

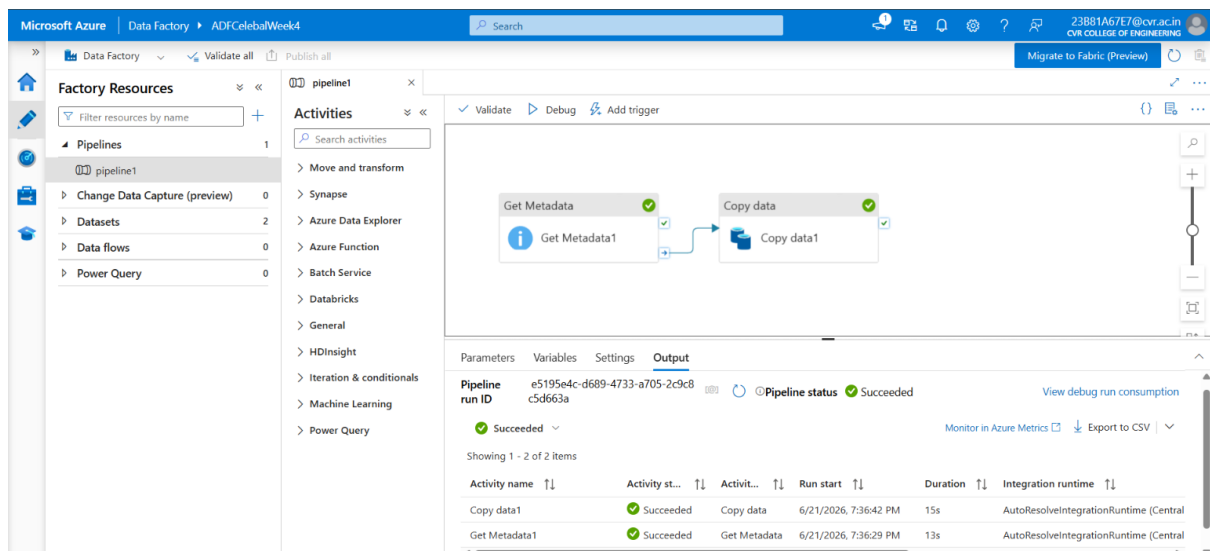


TASK 6: Build a pipeline using Copy Data activity and configure source and destination.



TASK 6 : Execute the pipeline using Debug/Trigger and monitor execution.

Screenshot :Successful Pipeline Execution Using Debug



TASK 7: Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage.

The screenshot shows the 'Access Control (IAM)' page for the 'adfcelebalstorage' storage account. The left sidebar lists various Azure services, with 'Access Control (IAM)' selected. The main area shows a table of role assignments. The table has columns for Name, Type, Role, Scope, and Condition. The roles listed are 'Owner' (2 assignments), 'Reader' (1 assignment), and 'Storage Blob Data Contributor' (1 assignment). The 'Reader' role is assigned to the 'ADFCelebaWeek4' managed identity. The 'Storage Blob Data Contributor' role is also assigned to the 'ADFCelebaWeek4' managed identity.

Name	Type	Role	Scope	Condition
Owner (2)				
DONTHULA LEKHANA 23B81A67E7@cvr.ac.in	User	Owner	Subscription (Inherited)	None
DONTHULA LEKHANA 23B81A67E7@cvr.ac.in	User	Owner	Subscription (Inherited)	None
Reader (1)				
ADFCelebaWeek4 538172ab-1961-469c-8...	Managed identity	Reader	This resource	None
Storage Blob Data Contributor (1)				
ADFCelebaWeek4 538172ab-1961-469c-8...	Managed identity	Storage Blob Data Contributor	This resource	Add

Output file successfully copied to the destination-data container after pipeline execution.

The screenshot shows the 'destination-data' container in the Microsoft Azure portal. The container is empty except for one file named 'Sample - Superstore.csv'. The file details are as follows:

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/21/2026, 7:43:16 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Brief Summary

- Explore Azure Portal and Created a Resource Group and Storage Account in Microsoft Azure.
- Created source and destination Blob Storage containers.
- Uploaded the Superstore CSV dataset to the source container.
- Created an Azure Data Factory and explored its Author, Monitor, and Manage sections.
- Configured a Linked Service to connect Azure Data Factory with Blob Storage.
- Created source and destination datasets for the pipeline.
- Built a pipeline using Get Metadata and Copy Data activities.
- Used Get Metadata to check the source file details before copying the data.
- Copied the dataset from the source container to the destination container using the Copy Data activity.
- Assigned the required IAM roles to enable access between Azure Data Factory and the Storage Account.
- Executed the pipeline successfully using Debug and monitored the execution status.
- Verified that the output file was available in the destination container after execution.
- Validated the successful transfer of data by checking both the pipeline results and the output file.
- Through this assignment, I learned how to create and manage a basic data pipeline in Azure Data Factory and gained practical experience with cloud storage, data movement, metadata validation, and access management.